Forecasting Rapeseed Volatility Changes with HAR-log and ElasticNet Models

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Abstract

We study the predictability of forward changes in log realised volatility for Euronext rapeseed futures using a Heterogeneous Autoregressive (HAR) model in logarithms and an ElasticNet regularised variant. Trained on a 5,773-day development set and validated on a 508-day holdout, both models produce statistically significant forecasts of volatility at 1, 5, 10 and 20-day horizons. HAR-log achieves directionl accuracy of 62%–72% across horizons; ElasticNet matches or slightly improves this.

1 Data

Daily OHLCV prices for the front-month Euronext rapeseed (MATIF) futures contract span approximately 25 years of trading history (6,281 trading days). An out-of-time holdout split at 20 April 2024 allocates 5,773 days for model development and 508 days for validation. An embargo equal to the forecast horizon is applied at the split boundary to prevent information leakage.

2 Definitions and Target Variable

Let $r_t = \log(C_t/C_{t-1})$ denote the daily log return on the front-month contract, where C_t is the closing price on day t.

Realised Volatility

We define realised volatility over a window $[a, b]$ as the sample standard deviation of daily log returns within that window:

$$\mathrm{RV}_{a:b} = \sqrt{\frac{1}{b-a}\sum_{i=a}^{b}\big(r_i - \bar{r}_{a:b}\big)^2}, \qquad \bar{r}_{a:b} = \frac{1}{b-a+1}\sum_{i=a}^{b} r_i.$$

This requires at least two observations ($b > a$). For the special case of a 1-day window, we use the absolute daily return as a proxy: $\mathrm{RV}_{t:t} = |r_t|$ [2]. Days with zero realised volatility are excluded before taking logarithms.

Forecast Target

The target variable is the forward change in log realised volatility:

$$y_{t,h} = \log \mathrm{RV}_{t+1:t+h} - \log \mathrm{RV}_{t-h+1:t}$$

i.e. future log volatility minus past log volatility over equal-length windows of h days. This difference-in-logs formulation is approximately stationary and centred near zero. We evaluate horizons $h \in \{1, 5, 10, 20\}$ days.

3 Model

We use a HAR-log specification following [1]: regression of $y_{t,h}$ on the log realised volatility at multiple aggregation windows, plus an intercept:

$$\hat{y}_{t,h} = \beta_0 + \beta_1 \log \mathrm{RV}_{t:t} + \beta_5 \log \mathrm{RV}_{t-4:t} + \beta_{10} \log \mathrm{RV}_{t-9:t} + \beta_{20} \log \mathrm{RV}_{t-19:t}.$$

Both models are fit with an expanding window: for each prediction at time t, all data from the start of the sample through $t-1$ is used for estimation (minimum 100 observations).

ElasticNet model adds combined ℓ_1/ℓ_2 regularisation, with the mixing parameter and penalty strength chosen by cross-validation within each expanding window.

Evaluation

All metrics are computed on *non-overlapping* observations to avoid inflated significance from return overlap: for horizon h, we subsample every h-th observation, compute the statistic, and average across all h phase offsets. We report phase-averaged Spearman rank correlation (IC) with approximate bootstrap 95% confidence intervals,[1] and directional accuracy (DA).

4 Results

4.1 Cross-Validation (Expanding Window)

Table 1: Expanding-window CV on the 5,773-day development set.

Horizon	Model	IC	DA (%)	R^2	n_{indep}
1d	HAR-log	0.63	72.50	0.44	5160
1d	ElasticNet	0.62	71.80	0.42	4018
5d	HAR-log	0.56	70.20	0.34	1116
5d	ElasticNet	0.61	72.00	0.42	646
10d	HAR-log	0.50	66.20	0.27	560
10d	ElasticNet	0.46	64.70	0.27	564
20d	HAR-log	0.42	62.40	0.22	281
20d	ElasticNet	0.55	68.50	0.37	281

All horizons show statistically significant IC for both models.

[1] Bootstrap CIs are computed by resampling non-overlapping observations within each phase. This removes the main source of dependence (overlapping returns) but may not fully account for slow-changing market conditions.

4.2 Out-of-Time Holdout

Table 2: Holdout performance (April 2024 onward, 508 trading days).

Horizon	Model	IC	DA (%)	R^2
1d	HAR-log	0.64	71.70	0.46
1d	ElasticNet	0.64	73.10	0.46
5d	HAR-log	0.61	71.10	0.35
5d	ElasticNet	0.61	63.50	0.04
10d	HAR-log	0.43	62.40	0.18
10d	ElasticNet	0.49	62.20	0.07
20d	HAR-log	0.56	71.40	0.29
20d	ElasticNet	0.57	59.00	0.20

All holdout ICs are positive and consistent with CV estimates and while directional accuracy of both models is well above 50%, HAR-log delivers more balanced performance across both ranking and regression metrics on the holdout data.

5 Naive Baselines

To verify that the HAR-log model adds value beyond trivial forecasts, we compare against two parameter-free baselines. *Vol mean-reversion* predicts $\log(\bar{\mathrm{RV}}/\mathrm{RV}_t)$, where $\bar{\mathrm{RV}}$ is the expanding-window mean—this captures the simplest possible mean-reversion signal. *Vol momentum* predicts $\Delta\mathrm{RV}_t$, testing whether the recent direction of vol changes persists.

Table 3: Directional accuracy (%) — HAR-log vs. naive baselines (dev set).

Horizon	HAR-log	Vol-MR	Vol-Mom
1d	72.50	71.20	35.10
5d	70.20	65.80	40.50
10d	66.20	63.10	48.70
20d	62.40	59.80	48.40

HAR-log consistently exceeds both baselines. Vol-momentum performs worse than a coin flip at all horizons, confirming that volatility changes mean-revert rather than trend. The margin over vol-MR is modest at 20d but substantial at shorter horizons, confirming that the multi-scale structure of HAR adds genuine predictive value beyond simple mean-reversion.

6 Feature Augmentation via Lasso

We tested whether additional features could improve on the bare HAR-log model by augmenting it with up to 155 candidate predictors—including cross-commodity spreads, positioning, macro indicators, and technicals—and applying LassoCV to select survivors.

The augmented model improves both IC and DA in cross-validation and on holdout (Table 4). However, in our preliminary monthly backtest (Section 7), the augmented signal underperformed bare HAR-log. One possible explanation is that the extra features have regime-dependent relationships with volatility, making the augmented forecast less stable day-to-day even when its

Table 4: HAR-log vs. HAR + Lasso survivors on the development set (expanding-window CV) and holdout.

	Model	CV IC	Holdout IC	Holdout DA (%)	Feats
10d	HAR-log	0.50	0.43	62.40	4
10d	HAR + Lasso	0.63	0.50	67.20	15
20d	HAR-log	0.42	0.56	71.40	4
20d	HAR + Lasso	0.67	0.53	64.70	15

average ranking quality is higher. A different trading implementation—for example, continuous position sizing rather than a binary monthly signal—might be better suited to exploit the augmented model's strengths.

The recurring Lasso survivors—crush spread, positioning concentration (HHI), fund share, inflation breakevens, and cross-commodity correlations—are statistically significant and economically interpretable. They may be useful as a confidence overlay for position sizing or as a monitoring tool for regime shifts, rather than as direct forecast inputs.

7 Preliminary Backtest

As a directional check on economic significance, we translate the 10-day HAR-log forecast into monthly-rebalanced futures positions. Two strategies are tested: *vol-timing sign* (long when vol predicted to fall, flat otherwise) and *contrarian-vol* (long when falling, short when rising).

Table 5: Backtest results, April 2024–May 2026 (508 trading days).†

Strategy	CumPnL (%)	Sharpe	MaxDD (%)	Ann. Turn
Vol-timing sign	19.88	0.66	−16.76	7.70
Contrarian-vol	25.85	0.64	−24.93	14.88
Buy-Hold	13.84	0.40	−20.07	0.49
Naive-Mom-10d	−40.73	−1.06	−46.90	61.76
Naive-MR-10d	12.20	0.37	−21.56	61.76

†Transaction cost of 12.5 bps per unit turnover. Approximately 24 monthly rebalances—a limited trading sample.

Both signal-based strategies outperform buy-and-hold and naive baselines over the holdout period. Performance is robust to cost assumptions: at 0 bps the gross Sharpe is 0.72 for both strategies; at 12.5 bps only 2–5% of cumulative PnL is lost to costs, owing to low monthly turnover.

8 Time-Series Foundation Models

As an additional benchmark we evaluated two pre-trained time-series foundation models (TSFMs) in a zero-shot setting on the same holdout period: Amazon's *Chronos-2* [3] (via forecasting log-volatility levels directly and calculating differences) and *Kronos* [4] (via a return-level forecast converted to realised volatility). Neither model was fine-tuned on rapeseed or commodity data. Kronos was trained on a large corpus of time-series data from various asset classes, whereas Chronos2 is a 120M parameter model trained on all forms of time-series data, including some financial data.

Both foundation models substantially underperform HAR-log at every horizon. Chronos2-direct produces worse-than-random results. Kronos-return retains decent directional accuracy

Table 6: Foundation-model holdout performance vs. HAR-log.

Horizon	Model	IC	DA (%)	R^2
1d	HAR-log	0.64	71.70	0.46
1d	Chronos2-direct	0.47	66.00	0.22
1d	Kronos-return	0.34	63.10	−0.41
5d	HAR-log	0.61	71.10	0.35
5d	Chronos2-direct	−0.37	40.20	−1.18
5d	Kronos-return	0.56	69.10	−0.55
10d	HAR-log	0.43	62.40	0.18
10d	Chronos2-direct	−0.27	44.70	−1.25
10d	Kronos-return	0.13	53.20	−1.13
20d	HAR-log	0.56	71.40	0.29
20d	Chronos2-direct	−0.42	38.60	−1.43
20d	Kronos-return	0.54	67.10	−0.73

but with strongly negative R^2. We tried varying context windows for both but found similar results.

9 Conclusion

A four-feature HAR-log model and its ElasticNet variant produce statistically significant out-of-sample forecasts of volatility changes for Euronext rapeseed futures across all horizons tested (1–20d). Lasso-selected auxiliary features (cross-commodity, positioning, macro) improve ranking quality and directional accuracy on holdout, but do not improve trading performance in our preliminary monthly backtest. The backtest provides encouraging but small-sample evidence that the 10-day signal carries economic value under realistic transaction costs.

10 Further Work

Since the HAR-log model uses only price-derived features, the same framework can be applied directly to other commodity futures—particularly more liquid markets such as crude oil, gold, or major grains—to test whether the volatility-change signal generalises across asset classes. It could also be interesting to explore if fine-tuned TSFMs models can challenge the HAR-log model.

References

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